# Adding the Comments Table

Lecture Notes · Game DB Junior

**In this lesson**

- What we have done so far
- What a **schema diagram** is
- Designing the **comments** table

## 1. Where we are

We set up a database for a photo sharing app. Right now it only has **users** and **photos**.

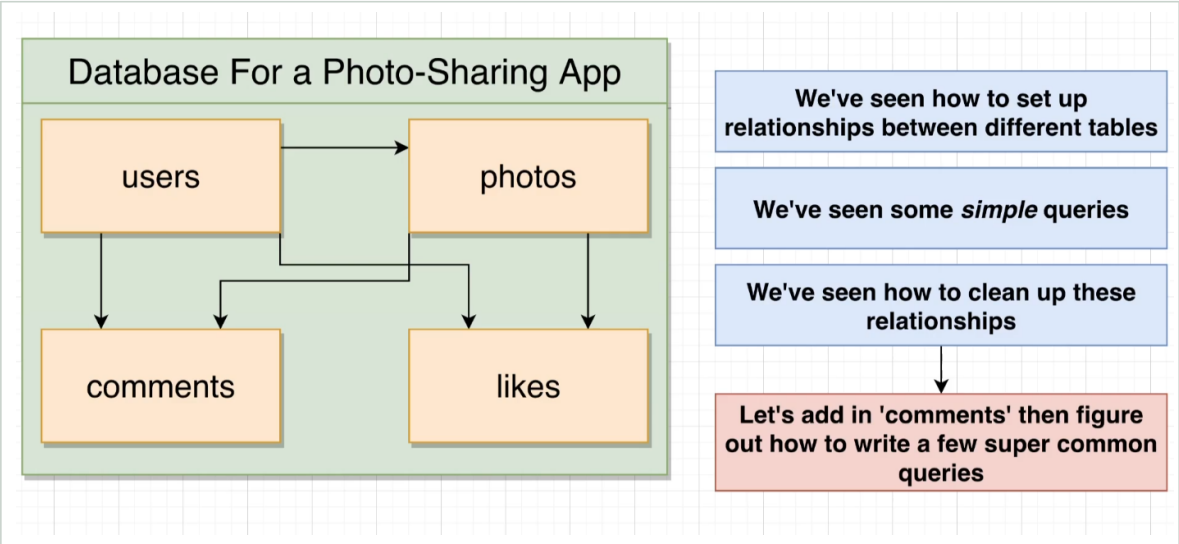



*Figure 1 — What we have done, and what comes next*

- We have seen how to set up a **relationship** between these two tables with foreign keys.
- We have seen some **simple queries**.
- We have seen a bit about keeping **data consistency** with the deletion constraints (ON DELETE).

## 2. What comes next

Now we add more complexity: a new table called **comments**. It is related to users and to photos. Then we learn how to write some **very common queries**, for example:

- Get all the comments created by a given user.
- Get a photo with all the comments on it.

Our earlier queries on related data were very simple. Now the focus is on more complicated queries.

## 3. The schema diagram

A **schema diagram** shows the **structure** of our database: the tables, their columns and the type of each column.

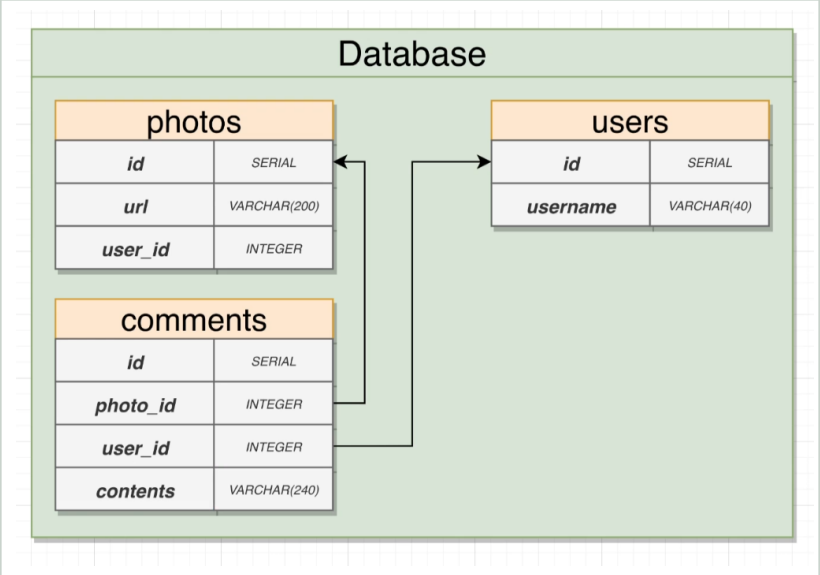



*Figure 2 — Schema diagram with photos, users and the new comments table*

## 4. Designing the comments table

Each comment must be related to a **photo** (which photo it is for) and to a **user** (who wrote it).

| Column | Type | What it stores |
|---|---|---|
| id | SERIAL | So we can refer to each single comment |
| photo_id | INTEGER | A reference to the id of the photo this comment is for |
| user_id | INTEGER | A reference to the user who created the comment |
| contents | VARCHAR(240) | The text of the comment. Up to 240 characters; longer text gives an error |

## 5. Summary

| Idea | Remember |
|---|---|
| So far | users and photos, a foreign key relationship, simple queries, ON DELETE |
| Schema diagram | Shows the structure of the database |
| comments | id, photo_id (points at photos), user_id (points at users), contents |

**Next lesson:** reset the database so everyone has the same structure and data, then create users and photos again and create comments for the first time.

**Sources:** Lecture transcript and lecture images (Figures 1–2)